Customers Like You

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About: The "Customers who bought this also bought" box and customer groups

Recommendation

Pilot Truncated SVD with 100 components for the "also bought" box. We tested it on purchases from 9 Oct 2011 to 9 Dec 2011, after the training period. For 23.2% of 1,913 customers with purchase history, at least one product they bought for the first time was among their 10 personalised suggestions, against 17.4% for best sellers: an estimated gain of 5.8 points (95% interval 3.4 to 8.3). PCA scored 22.9%, and this test does not establish a difference between the two.

Method, size and speed

Each cell of the customer x product table is log(1 + total quantity), and products need at least 10 buyers before the test period, leaving 97.9% of cells empty. SVD compresses this table as it is. PCA instead describes differences from the average customer. We fixed 100 components before testing, from a curve on training data only. The full table needs 161.4 MB and sparse matrix storage 4.99 MB. The model's customer and product tables need 7.33 MB, 22 times less than the full table but more than sparse storage. From the prepared purchase table, fitting and finding the 10 most similar products for all 3,687 products took 0.707 s on an Apple M4 Mac (10 logical CPUs). Raw-data cleaning is extra, and production times will differ.

What compression loses

Rebuilt from 100 components, the log-transformed table has a relative reconstruction error of 72.7% (Frobenius norm). It measures how roughly the model approximates purchase patterns, not a share of purchases or quantities kept. Suggestions follow broad patterns and may miss niche items.

Customer groups for marketing

Four groups from the first two model components, named from their products:

- **High-volume wholesalers: children's ranges**: 1,365 customers (25%), e.g. Dolly girl beaker.
- **Mid-size regular buyers: home decor**: 1,910 customers (35%), e.g. White pearl beaded heart, small.
- **Smaller regular buyers: children's ranges, party & paper goods**: 1,387 customers (25%), e.g. Nurse's bag soft toy.
- **Occasional small buyers: mixed gifts**: 811 customers (15%), e.g. Antique silver tea glass engraved.

All groups are mainly UK-based (85.5% to 97.7% of customers); some have a larger overseas share. Treat the groups as campaign ideas to test.

Risks and next steps

- Coverage: 1,913 of 2,308 test customers (82.9%) had history and a catalogue product to score. 2,555 of 2,849 distinct test products (89.7%) are in the catalogue, and 70.7% of new customer-product pairs could be scored.
- Product-page lists work for any visitor on a catalogue product, new customers included. Personalised suggestions need purchase history, and new products or those with under 10 buyers need best sellers as a fallback.
- The offline hit rate does not prove extra sales. Next: an A/B pilot against best sellers measuring clicks and orders, a marketing review of the group names, and weekly coverage tracking.

Data: Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D. Licence CC BY 4.0.